1. Executive Summary

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1. Executive summary
- Train games=940 | Derby games=75
- Nested CV=5 outer folds x 3 inner folds (expanding time-aware)
- Selected model family: Regime-aware simplex stack (Ridge/Huber/HistGB/GBR/XGB/HistGB-bag) + q50 blend
- Selected Elo variant: elo_dynamic_k_teamha_decay
- Nested outer RMSE/MAE: 37.971 / 30.063
- Selected postprocess module/eta: expand_affine_global / 0.55
- Calibration=none | Scale=none | Regime stack=False
- Runtime safety: fast_mode=False | budget_triggered=none
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1B. Runtime Bottleneck And Fixes

Runtime bottleneck and fixes
- Profile-first instrumentation added for hotspot phase totals, per-family _predict_family timing, model fit counts, and top slow events.
- Shared split-level matrix/weight cache keyed by train/val index signatures + feature signature + half-life to eliminate repeated _xy/time-weight prep.
- Bundled family prediction path prepares X/y/w once per split and fits Ridge/Huber/HistGB/GBR/XGB/HistGB-bag/q50 sequentially.
- HistGB bagging reduced to 4 models by default (fast mode uses 1), with deterministic seed schedule and budget-aware auto-reduction.
- Candidate scan uses conservative early pruning: coarse score on first inner split, then full evaluation on top candidates only.
- Budget enforcement: total time, scan time, and fit cap stop further scanning/training and continue with best-so-far.
- Budgets triggered: none
- Fast mode=False; optimization toggles enabled=True.
- Reproducibility: seed=23, OMP_NUM_THREADS=1, MKL_NUM_THREADS=1.
- Artifact validation: predictions rows=75 (expected 75), rankings rows=165 (expected 165), ranks unique 1..165=True.

1D. Runtime Focus Phase Timing Summary (1/1)

phase	total_sec	calls
_predict_family	1624.179	7174
_scan_and_select_core_candidates	880.585	6
_core_scan_score	880.524	1810
_generate_inner_oof_and_outer_preds	740.003	42
_prepare_variant_outer_data	1.277	6
_build_split_tables	1.159	24

1E. _predict_family Aggregated Timing / Fit Counts (1/1)

family	predict_time_sec	predict_calls	fit_count
histgb	921.910	3079	3079
gbr	267.273	169	169
histgb_bag	193.709	169	676
histgb_q50	104.436	169	169
xgb	91.988	169	169
huber	39.921	169	169
ridge	3.864	3079	3079
histgb_q20	0.461	1	1
histgb_q80	0.455	1	1
enet_f2	0.162	169	169

2-4. Changes / Leakage / Nested CV

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2. What changed vs previous pipeline
- Nested tuning for post-model layers and key hyperparameters (Ridge alpha, HistGB grid, recency half-life).
- Regime-specific simplex stacking + regime-specific calibration + regime/pool scale correction.
- Recency features: EMA multi-alpha, last3/5/8, trend, consistency ratio.
- Upgraded Elo variants: dynamic-K, team-specific regularized home effect, inactivity decay.
- HistGB seed bagging and quantile HistGB (median, plus q20/q80 for diagnostics).
- Shift diagnostics (KS + PSI + out-of-range fraction) and OOD-triggered mitigation.

3. Data constraints and leakage controls
- All features are pregame-only and built sequentially.
- Static rating models refit inside each train split; no outer-fold leakage in postprocessing.
- Derby predictions are neutral-site (home effects zeroed for derby features).

4. Nested CV design (outer/inner schematic)
 outer_fold=1 train_n=125 val_n=193 train_end=2025-01-12 val=2025-01-23..2025-02-19
 outer_fold=2 train_n=318 val_n=205 train_end=2025-02-19 val=2025-03-03..2025-04-02
 outer_fold=3 train_n=523 val_n=151 train_end=2025-04-02 val=2025-04-14..2025-04-30
 outer_fold=4 train_n=674 val_n=102 train_end=2025-04-30 val=2025-05-13..2025-05-29
 outer_fold=5 train_n=776 val_n=164 train_end=2025-05-29 val=2025-05-30..2025-06-30
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5. Outer-fold selections and metrics (1/1)

| outer_candidate_id | dataset | feature_set | n_folds | edge_threshold | aggregation | stacking | calibration | calibrator | oof_roc_auc | oof_pr_auc | oof_brier | oof_ece | oof_mcc | oof_f1 | oof_precision | oof_recall | oof_threshold | oof_prevalence | oof_n_samples | oof_n_positives | oof_n_negatives | outer_roc_auc | outer_pr_auc | outer_brier | outer_ece | outer_mcc | outer_f1 | outer_precision | outer_recall | outer_threshold | outer_prevalence | outer_n_samples | outer_n_positives | outer_n_negatives | delta_roc_auc | delta_pr_auc | delta_brier | delta_ece | delta_mcc | delta_f1 | delta_precision | delta_recall | delta_threshold | delta_prevalence | delta_n_samples | delta_n_positives | delta_n_negatives | n_features | n_edges | n_nodes | mean_degree | density | weakly_connected_components | largest_wcc_size | largest_wcc_frac | singleton_frac | isolate_frac | mean_in_deg | mean_out_deg | max_in_deg | max_out_deg | gini_in_deg | gini_out_deg | frac_nodes_deg_ge_2 | frac_nodes_deg_ge_5 | frac_nodes_deg_ge_10 | reciprocity | mean_edge_weight | median_edge_weight | frac_edges_weight_ge_0.5 | frac_edges_weight_ge_0.9 | frac_edges_weight_ge_0.99 | mean_node_strength | median_node_strength | max_node_strength | gini_node_strength | frac_nodes_strength_ge_1 | frac_nodes_strength_ge_5 | frac_nodes_strength_ge_10 | mean_in_strength | mean_out_strength | gini_in_strength | gini_out_strength | frac_nodes_with_in_strength | frac_nodes_with_out_strength | frac_nodes_with_both_strength | mean_abs_log_in_out_strength_ratio | frac_nodes_abs_log_ratio_ge_0.5 | frac_nodes_abs_log_ratio_ge_1.0 | frac_nodes_abs_log_ratio_ge_2.0 | mean_abs_log_in_out_strength_ratio_ge_0.5 | mean_abs_log_in_out_strength_ratio_ge_1.0 | mean_abs_log_in_out_strength_ratio_ge_2.0 | mean_abs_log_in_out_strength_ratio_ge_0.5_by_mean_node_s_cap_ |
|---|---|---

5. Outer-fold RMSE/MAE table (1/1)

outer_fold	rmse	mae	bias	pred_std	actual_std	resid_std	tail_rmse	xtail_rmse	tail_bias	tail_dispersion_rat	corr
1	44.657	34.939	-7.256	40.979	45.098	44.063	62.824	79.116	-13.051	0.721	0.479
2	39.693	30.810	-0.823	29.011	44.241	39.684	61.122	69.314	-16.989	0.495	0.477
3	36.095	29.307	3.712	26.748	38.591	35.904	52.674	55.995	-1.165	0.397	0.444
4	31.300	25.092	3.663	28.324	40.496	31.085	47.264	50.880	-2.245	0.510	0.643
5	38.112	30.166	0.288	25.928	47.133	38.111	63.016	70.033	-10.116	0.388	0.590

6. Feature additions and recency ablations (core scan) (1/1)

feature_profile	half_life_days	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
signal_core		42.422	43.079	50.369	345
signal_core	45.000	42.464	43.095	50.444	345
signal_core	30.000	42.485	43.105	50.472	344
signal_core	20.000	42.501	43.116	50.500	346
signal_core	10.000	42.683	43.253	50.798	340

6. Core scan results (top rows) (1/4)

candidate_ke	lo_variant	feature_prof	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber	scan_ridge_s	scan_stage_	pred_std_sca	scan_splits_u	scan_stage	outer_fold
ecay_signal	dynamic_k_tea	signal_core		8.000	0	34.446	37.460	29.326					36.772	28.990	-14.204	3	full	5
ecay_signal	dynamic_k_tea	signal_core		4.000	0	34.523	37.460	29.326					36.993	29.116	-14.204	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	8.000	0	34.573	37.656	29.467					36.844	29.019	-14.134	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	4.000	0	34.625	37.656	29.467					36.995	29.099	-14.134	3	full	5
ecay_signal	dynamic_k_tea	signal_core		8.000	2	34.632	37.832	29.297					36.772	28.990	-13.994	3	full	5
ecay_signal	dynamic_k_tea	signal_core		8.000	4	34.644	37.840	29.597					36.772	28.990	-14.153	3	full	5
ecay_signal	dynamic_k_tea	signal_core		2.000	0	34.667	37.460	29.326					37.405	29.426	-14.204	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	8.000	2	34.701	37.883	29.627					36.844	29.019	-13.954	3	full	5
ecay_signal	dynamic_k_tea	signal_core		4.000	2	34.709	37.832	29.297					36.993	29.116	-13.994	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	8.000	4	34.713	37.910	29.677					36.844	29.019	-14.060	3	full	5
ecay_signal	dynamic_k_tea	signal_core	30.000	8.000	0	34.718	37.831	29.686					37.000	29.087	-14.100	3	full	5
ecay_signal	dynamic_k_tea	signal_core		4.000	4	34.721	37.840	29.597					36.993	29.116	-14.153	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	2.000	0	34.735	37.656	29.467					37.309	29.281	-14.134	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	4.000	2	34.754	37.883	29.627					36.995	29.099	-13.954	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	4.000	4	34.765	37.910	29.677					36.995	29.099	-14.060	3	full	5
ecay_signal	dynamic_k_tea	signal_core	30.000	4.000	0	34.766	37.831	29.686					37.137	29.168	-14.100	3	full	5
ecay_signal	dynamic_k_tea	signal_core		8.000	1	34.827	38.228	29.620					36.772	28.990	-13.695	3	full	5
ecay_signal	dynamic_k_tea	signal_core		2.000	2	34.854	37.832	29.297					37.405	29.426	-13.994	3	full	5
ecay_signal	dynamic_k_tea	signal_core	30.000	2.000	0	34.859	37.831	29.686					37.405	29.332	-14.100	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	2.000	2	34.864	37.883	29.627					37.309	29.281	-13.954	3	full	5
ecay_signal	dynamic_k_tea	signal_core		8.000	3	34.864	38.320	30.018					36.772	28.990	-13.241	3	full	5
ecay_signal	dynamic_k_tea	signal_core		2.000	4	34.865	37.840	29.597					37.405	29.426	-14.153	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	2.000	4	34.875	37.910	29.677					37.309	29.281	-14.060	3	full	5
ecay_signal	dynamic_k_tea	signal_core	45.000	8.000	1	34.877	38.240	29.975					36.844	29.019	-13.798	3	full	5

6. Core scan results (top rows) (2/4)

candidate_ke	lo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_s	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber_s	scan_ridge_s	scan_stage	pred_std_sca	scan_splits_u	scan_stage	outer_fold
decay_slow	dynamic_k_tea	signal_core	30.000	8.000	4	34.890	38.168	29.905					37.000	29.087	-13.968	3	full	5
ecay_slow	dynamic_k_tea	signal_core		1.000	0	34.897	37.460	29.326					38.060	29.995	-14.204	3	full	5
ecay_slow	dynamic_k_tea	signal_core		4.000	1	34.905	38.228	29.620					36.993	29.116	-13.695	3	full	5
decay_slow	dynamic_k_tea	signal_core	20.000	8.000	0	34.905	38.039	29.958					37.245	29.236	-13.914	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	8.000	2	34.913	38.202	29.955					37.000	29.087	-13.841	3	full	5
decay_slow	dynamic_k_tea	signal_core	45.000	4.000	1	34.930	38.240	29.975					36.995	29.099	-13.798	3	full	5
decay_slow	dynamic_k_tea	signal_core	45.000	1.000	0	34.930	37.656	29.467					37.865	29.748	-14.134	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	4.000	4	34.937	38.168	29.905					37.137	29.168	-13.968	3	full	5
ecay_slow	dynamic_k_tea	signal_core		4.000	3	34.941	38.320	30.018					36.993	29.116	-13.241	3	full	5
decay_slow	dynamic_k_tea	signal_core	20.000	4.000	0	34.956	38.039	29.958					37.388	29.327	-13.914	3	full	5
decay_slow	dynamic_k_tea	signal_core	45.000	8.000	3	34.961	38.406	29.952					36.844	29.019	-13.542	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	4.000	2	34.961	38.202	29.955					37.137	29.168	-13.841	3	full	5
decay_slow	dynamic_k_tea	signal_core	20.000	8.000	4	34.993	38.201	30.054					37.245	29.236	-13.984	3	full	5
decay_slow	dynamic_k_tea	signal_core	45.000	4.000	3	35.013	38.406	29.952					36.995	29.099	-13.542	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	8.000	3	35.030	38.439	30.181					37.000	29.087	-13.379	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	2.000	4	35.031	38.168	29.905					37.405	29.332	-13.968	3	full	5
decay_slow	dynamic_k_tea	signal_core	20.000	2.000	0	35.033	38.039	29.958					37.609	29.462	-13.914	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	1.000	0	35.033	37.831	29.686					37.902	29.729	-14.100	3	full	5
decay_slow	dynamic_k_tea	signal_core	45.000	2.000	1	35.040	38.240	29.975					37.309	29.281	-13.798	3	full	5
decay_slow	dynamic_k_tea	signal_core	20.000	4.000	4	35.043	38.201	30.054					37.388	29.327	-13.984	3	full	5
ecay_slow	dynamic_k_tea	signal_core		2.000	1	35.049	38.228	29.620					37.405	29.426	-13.695	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	8.000	1	35.054	38.460	30.141					37.000	29.087	-14.088	3	full	5
decay_slow	dynamic_k_tea	signal_core	30.000	2.000	2	35.055	38.202	29.955					37.405	29.332	-13.841	3	full	5
decay_slow	dynamic_k_tea	signal_core	45.000	1.000	4	35.070	37.910	29.677					37.865	29.748	-14.060	3	full	5

6. Core scan results (top rows) (3/4)

candidate_ke	lo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_s	scan_gbr_sc	scan_gbr_sc	scan_huber_	scan_huber_s	scan_ridge_s	scan_stagepe	ned_std_sca	scan_splits_u	scan_stage	outer_fold
decay_sig	dynamic_k_tea	signal_core	30.000	4.000	3	35.078	38.439	30.181					37.137	29.168	-13.379	3	full	5
decay_sig	dynamic_k_tea	signal_core		2.000	3	35.086	38.320	30.018					37.405	29.426	-13.241	3	full	5
decay_sig	dynamic_k_tea	signal_core		1.000	4	35.094	37.840	29.597					38.060	29.995	-14.153	3	full	5
decay_sig	dynamic_k_tea	signal_core	10.000	8.000	0	35.097	38.135	29.982					37.696	29.529	-13.410	3	full	5
decay_sig	dynamic_k_tea	signal_core	30.000	4.000	1	35.101	38.460	30.141					37.137	29.168	-14.088	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	2.000	4	35.120	38.201	30.054					37.609	29.462	-13.984	3	full	5
decay_sig	dynamic_k_tea	signal_core	45.000	2.000	3	35.123	38.406	29.952					37.309	29.281	-13.542	3	full	5
decay_sig	dynamic_k_tea	signal_core	10.000	4.000	0	35.154	38.135	29.982					37.859	29.643	-13.410	3	full	5
decay_sig	dynamic_k_tea	signal_core	30.000	2.000	3	35.172	38.439	30.181					37.405	29.332	-13.379	3	full	5
decay_sig	dynamic_k_tea	signal_core	10.000	8.000	4	35.173	38.291	30.197					37.696	29.529	-13.279	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	1.000	0	35.178	38.039	29.958					38.025	29.766	-13.914	3	full	5
decay_sig	dynamic_k_tea	signal_core	30.000	2.000	1	35.195	38.460	30.141					37.405	29.332	-14.088	3	full	5
decay_sig	dynamic_k_tea	signal_core	10.000	2.000	0	35.201	38.135	29.982					37.991	29.730	-13.410	3	full	5
decay_sig	dynamic_k_tea	signal_core	30.000	1.000	4	35.205	38.168	29.905					37.902	29.729	-13.968	3	full	5
decay_sig	dynamic_k_tea	signal_core	10.000	4.000	4	35.230	38.291	30.197					37.859	29.643	-13.279	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	8.000	3	35.251	38.740	30.574					37.245	29.236	-13.053	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	1.000	4	35.266	38.201	30.054					38.025	29.766	-13.984	3	full	5
decay_sig	dynamic_k_tea	signal_core	10.000	2.000	4	35.276	38.291	30.197					37.991	29.730	-13.279	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	4.000	3	35.301	38.740	30.574					37.388	29.327	-13.053	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	8.000	2	35.301	38.822	30.481					37.245	29.236	-13.421	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	4.000	2	35.351	38.822	30.481					37.388	29.327	-13.421	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	8.000	1	35.365	38.949	30.509					37.245	29.236	-13.710	3	full	5
decay_sig	dynamic_k_tea	signal_core	10.000	8.000	2	35.366	38.717	30.464					37.696	29.529	-12.800	3	full	5
decay_sig	dynamic_k_tea	signal_core	20.000	2.000	3	35.378	38.740	30.574					37.609	29.462	-13.053	3	full	5

6. Core scan results (top rows) (4/4)

candidate_k	elo_variant	feature_profile	half_life_days	ridge_alpha	histgb_idx	scan_score	scan_histgb	scan_histgb_s	scan_gbr_rank	scan_gbr_s	scan_huber_rank	scan_huber_s	scan_ridge_rank	scan_ridge_s	scan_stagger_ned_std_gap	scan_splits_used	scan_stage	outer_fold
decay_dynamic_k_tean	signal_core		20.000	4.000	1	35.415	38.949	30.509					37.388	29.327	-13.710	3	full	5
decay_dynamic_k_tean	signal_core		10.000	4.000	2	35.423	38.717	30.464					37.859	29.643	-12.800	3	full	5
decay_dynamic_k_tean	signal_core		20.000	2.000	2	35.428	38.822	30.481					37.609	29.462	-13.421	3	full	5
decay_dynamic_k_tean	signal_core		10.000	8.000	1	35.476	38.924	30.541					37.696	29.529	-12.894	3	full	5
decay_dynamic_k_tean	signal_core		20.000	2.000	1	35.492	38.949	30.509					37.609	29.462	-13.710	3	full	5
decay_dynamic_k_tean	signal_core		10.000	4.000	1	35.533	38.924	30.541					37.859	29.643	-12.894	3	full	5
decay_dynamic_k_tean	signal_core		10.000	2.000	1	35.580	38.924	30.541					37.991	29.730	-12.894	3	full	5
decay_dynamic_k_tean	signal_core		20.000	1.000	1	35.638	38.949	30.509					38.025	29.766	-13.710	3	full	5

7. Elo variant comparisons (1/1)

elo_variant	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
elo_dynamic_k_teamha_decay	42.510	43.129	50.516	1720

7. HistGB hyperparameter sweep summary (1/1)

histgb_idx	scan_score_mean	scan_histgb_rmse_mean	n	params
3	41.157	41.658	220	{'learning_rate': 0.025, 'max_depth': 5, 'max_leaf_nodes': 63, 'min_samples_leaf': 5
2	41.705	42.025	234	{'learning_rate': 0.032, 'max_depth': 4, 'max_leaf_nodes': 63, 'min_samples_leaf': 6,
7	42.346	42.066	134	{'learning_rate': 0.08, 'max_depth': 4, 'max_leaf_nodes': 127, 'min_samples_leaf': 5
4	41.530	42.285	200	{'learning_rate': 0.06, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 12
5	43.187	42.974	127	{'learning_rate': 0.07, 'max_depth': 5, 'max_leaf_nodes': 63, 'min_samples_leaf': 4,
1	41.897	43.124	210	{'learning_rate': 0.05, 'max_depth': 4, 'max_leaf_nodes': 31, 'min_samples_leaf': 8,
8	43.778	43.622	125	{'learning_rate': 0.045, 'max_depth': 7, 'max_leaf_nodes': 127, 'min_samples_leaf': 2
6	43.820	43.656	136	{'learning_rate': 0.05, 'max_depth': 6, 'max_leaf_nodes': 127, 'min_samples_leaf': 3
9	43.800	43.657	124	{'learning_rate': 0.065, 'max_depth': 6, 'max_leaf_nodes': 255, 'min_samples_leaf': 2
0	43.705	46.538	210	{'learning_rate': 0.04, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10

8. Regime-specific stacking / calibration results (1/1)

module_name	use_regime_sta	calibration_mod	scale_mode	eta	outer_rmse_mo	outer_mae_mo	outer_tail_dispersion	outer_tail_dispersion	outer_tail_bias_m	outer_corr_mean	outer_dispersion	regime_gain	gain_vs_poole	n
xpand_affine_gl	False	none	none	0.550	37.887	30.112	56.873	0.505	-9.101	0.528	0.707	-0.032		39
xpand_affine_gl	False	none	none	0.600	38.049	30.131	62.727	0.394	-10.234	0.591	0.560	0.000		1

9. Dispersion and scale correction analysis (1/1)

outer_fold	stage	pred_std	actual_std	rmse
1	raw_mean	24.836	45.098	40.174
1	calibrated	24.836	45.098	40.174
1	scaled	24.836	45.098	40.174
1	final	40.979	45.098	44.657
2	raw_mean	29.011	44.241	39.808
2	calibrated	29.011	44.241	39.808
2	scaled	29.011	44.241	39.808
2	final	29.011	44.241	39.693
3	raw_mean	26.748	38.591	35.957
3	calibrated	26.748	38.591	35.957
3	scaled	26.748	38.591	35.957
3	final	26.748	38.591	36.095
4	raw_mean	28.324	40.496	31.108
4	calibrated	28.324	40.496	31.108
4	scaled	28.324	40.496	31.108
4	final	28.324	40.496	31.300
5	raw_mean	25.928	47.133	38.117
5	calibrated	25.928	47.133	38.117
5	scaled	25.928	47.133	38.117
5	final	25.928	47.133	38.112

10. Shift diagnostics: Train recent vs Derby (1/1)

feature	train_recent_mean	derby_mean	train_recent_std	derby_std	ks_stat	ks_pvalue	psi	derby_outside_train_1_9	ood_flag
games_played_min	9.122	11.067	1.095	0.660	0.730	0.000	8.252	0.227	True
elo_diff_pre	65.009	-1.307	45.491	37.172	0.616	0.000	5.372	0.173	True
elo_neutral_diff_pre	1.489	-1.307	35.574	37.172	0.094	0.659	0.211	0.040	True
same_conf_flag	0.697	0.013	0.459	0.115	0.684	0.000	0.000	0.000	True
massey_diff	5.872	2.195	19.548	15.187	0.166	0.075	0.192	0.000	False
consistency_ratio_di	0.059	-0.008	0.657	0.577	0.113	0.427	0.167	0.013	False
_margin_last5_vs_sea	-1.434	1.554	16.927	19.606	0.105	0.514	0.133	0.067	False
volatility_sum	71.164	73.217	11.169	9.954	0.139	0.196	0.132	0.027	False
offdef_net_diff	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False
offdef_margin_neutr	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False

10. Shift Mitigation Notes

```
10. Shift mitigation rules
- Detected 4 OOD-like feature shifts (KS/PSI/out-of-range checks).
- Applied softer scaling (blend toward 1.0) and wider winsor bounds under OOD risk.
- No anti-compression guard applied (derby std 25.27, ref 27.77).
```

8. Regime weights (outer_fold_1_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	103	0.167	0.167	0.167	0.167	0.167	0.167

8. Regime weights (outer_fold_2_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	284	0.030	0.000	0.720	0.000	0.000	0.250

8. Regime weights (outer_fold_3_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	448	0.117	0.242	0.226	0.099	0.241	0.076

8. Regime weights (outer_fold_4_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	549	0.213	0.071	0.543	0.000	0.071	0.102

8. Regime weights (outer_fold_5_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	589	0.485	0.000	0.000	0.000	0.000	0.515

8. Calibration params (outer_fold_1_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	103	0.000	1.000

8. Calibration params (outer_fold_2_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	284	0.000	1.000

8. Calibration params (outer_fold_3_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	448	0.000	1.000

8. Calibration params (outer_fold_4_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	549	0.000	1.000

8. Calibration params (outer_fold_5_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	589	0.000	1.000

9. Scale params (outer_fold_1_selected) (1/1)

level	regime	n	scale
pooled	ALL	103	1.250

9. Scale params (outer_fold_2_selected) (1/1)

level	regime	n	scale
pooled	ALL	284	0.800

9. Scale params (outer_fold_3_selected) (1/1)

level	regime	n	scale
pooled	ALL	448	0.800

9. Scale params (outer_fold_4_selected) (1/1)

level	regime	n	scale
pooled	ALL	549	0.800

9. Scale params (outer_fold_5_selected) (1/1)

level	regime	n	scale
pooled	ALL	589	0.850

11. Residual Diagnostics (Summary)

```
11. Residual diagnostics
- OOF final RMSE=39.049, MAE=30.664, Bias=-0.721
- Residual std=39.042, skew=0.193, kurtosis=0.371
- Mean-stack RMSE=37.866
- Quantile median (q50) RMSE=37.488
- Mean+median blend RMSE=37.866
- Calibrated RMSE=37.866
- Tail error threshold (top 10%)=63.772
```

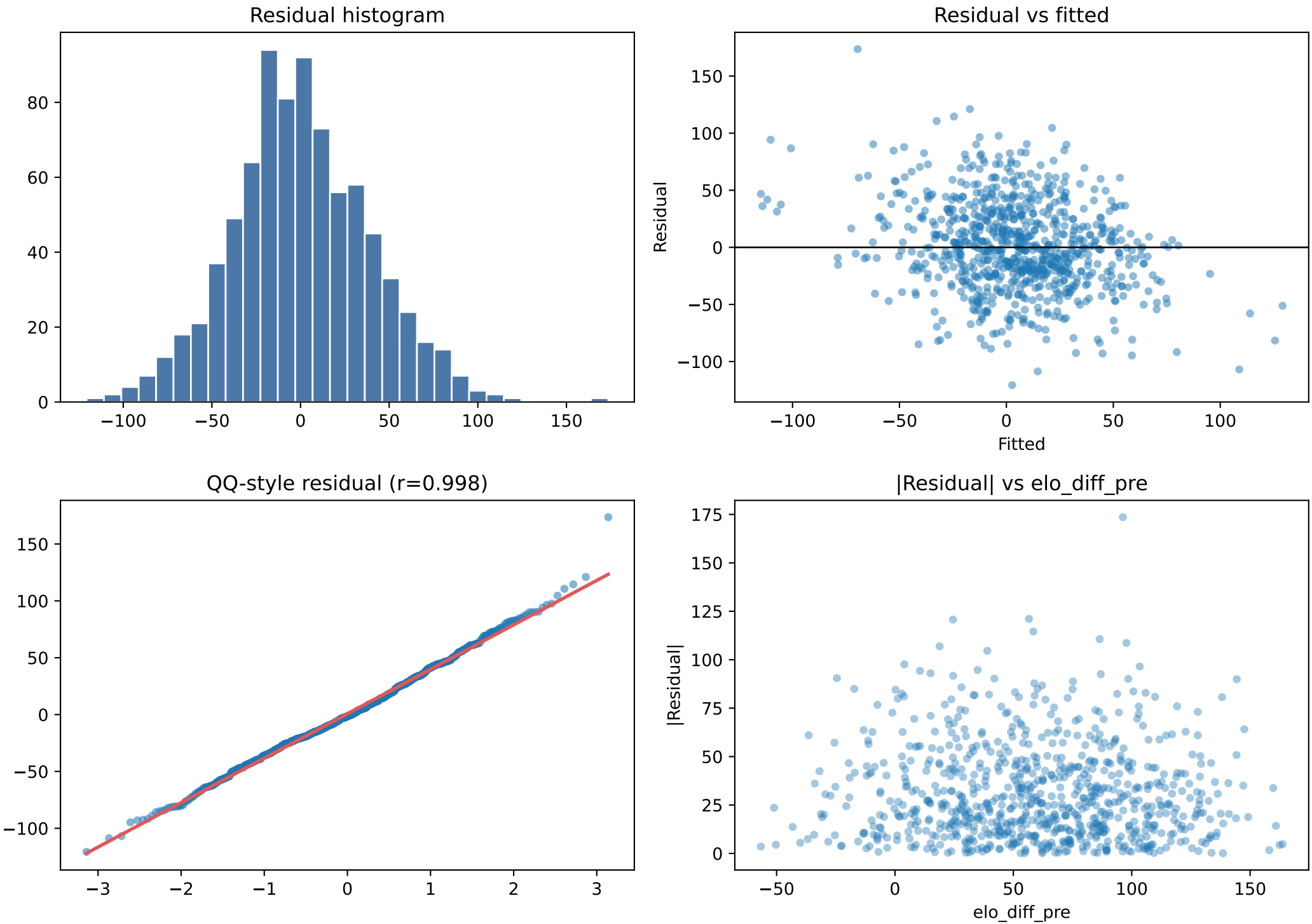
11. Tail error diagnostics (top 10% abs-error) (1/2)

Date	ret_diff	asset_ret	vol_ret	volatility	gain	gap	gap_pct	dow	month	month_end	month_start	quarter_end	quarter_start	Stency	row_indy	trend	pred_epe	pred_low	pred_high	pred_gap	pred_x	histgr	pred_obs	eff_ratio	prev_pred	prev_cap	pad_score	model	fraction	process	abs_error
2-02-19	0	96.284	0.396	12.252	3.792	6.949	3.000	3.000	1.000	0.000	0.000	0.000	0.000	0.499	304	104.00	43.45	-64.30	-11.33	-34.24	-40.15	-34.00	-34.24	-34.00	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	173.574
5-05-30	0	66.595	-6.552	16.212	2.572	9.441	8.000	9.000	1.000	0.000	-12.06	-11.87	1.381	804	104.00	-15.23	-18.04	-15.39	-22.40	-25.29	-13.11	-22.40	-20.78	0.00	5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	121.075
2-02-05	0	24.510	-5.428	-5.714	5.000	2.754	2.000	3.000	1.000	0.000	0.000	0.000	-1.387	236	118.00	-10.88	-15.80	-20.20	-5.341	-0.396	-2.798	-5.341	-2.735	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	120.705
3-03-17	0	68.435	-8.498	-15.581	5.462	8.262	5.000	4.000	1.000	0.000	0.000	0.000	2.786	393	90.00	-21.69	-35.57	-28.47	-22.27	-24.04	-26.76	-22.27	-29.85	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	114.545
2-02-19	0	86.464	-5.080	-3.786	1.085	7.720	3.000	3.000	1.000	0.000	0.000	0.000	0.567	275	78.00	-31.41	-38.08	-9.299	-15.06	-14.91	-1.523	-15.06	-9.176	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	110.623
4-04-02	0	97.770	-5.403	-0.800	2.003	0.529	4.000	6.000	1.000	0.000	-11.20	-11.196	-0.191	491	-94.00	-15.067	-3.007	-17.612	-14.502	-17.567	-9.845	-14.502	-12.410	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	108.654
2-02-05	0	18.858	-20.138	-16.536	3.036	7.000	3.000	2.000	0.000	1.000	0.000	0.000	-0.505	200	-2.00	-63.852	-30.532	-30.977	-68.266	-61.716	-65.944	-68.266	-76.630	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	106.906
2-02-19	0	88.985	-12.256	-6.286	3.893	1.004	4.000	2.000	1.000	0.000	0.000	0.000	1.510	269	126.00	-7.814	-34.062	-5.890	-12.328	-4.932	-8.900	-12.328	-9.926	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	104.606
4-04-02	0	3.955	-4.612	-2.442	1.782	-2.464	5.000	5.000	0.000	1.000	0.000	0.000	1.407	495	-94.00	-9.715	-18.352	-9.798	-4.160	-2.235	-7.146	-4.160	-1.416	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	97.598
1-01-24	0	03.358	-8.866	-3.893	5.004	-5.000	2.000	1.000	0.000	1.000	0.000	0.000	-0.726	165	-84.00	-20.08	-35.79	-16.831	-1.146	-1.783	-8.924	-1.146	-0.750	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	96.496
4-04-16	0	84.920	-9.480	-0.667	7.215	-2.564	8.000	8.000	0.000	1.000	-25.95	-25.86	0.799	591	-36.00	-37.049	-15.143	-5.406	-9.606	-9.547	-0.705	-9.606	-0.700	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	94.702
2-02-05	0	10.560	-6.613	-8.007	7.000	-9.318	2.000	3.000	1.000	0.000	0.000	0.000	0.294	223	-16.00	-21.34	-21.91	-5.414	-1.024	-9.612	-6.879	-1.024	-9.022	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	94.239
4-04-02	0	15.035	-3.581	-4.156	8.085	-7.196	6.000	5.000	1.000	0.000	-1.533	-1.781	0.342	479	-48.00	-41.835	-4.946	-7.736	-7.714	-5.382	-1.669	-7.714	-0.100	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	92.981
1-01-24	0	86.925	-4.124	-7.107	5.000	-7.000	1.000	2.000	0.000	1.000	0.000	0.000	-1.690	159	-60.00	-10.920	-0.209	-1.602	-8.573	-7.642	-5.272	-8.573	-3.601	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	92.507
2-02-19	0	24.570	-8.279	-3.173	7.550	-6.000	4.000	2.000	1.000	0.000	0.000	0.000	-0.097	316	-12.00	-35.909	-1.436	-1.394	-8.764	-2.624	-4.934	-8.764	-0.265	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	91.712
3-03-04	0	24.545	-5.403	-0.022	5.281	-5.020	4.000	4.000	1.000	0.000	0.000	0.000	-0.107	385	-100.00	-11.894	-2.382	-1.597	-5.409	-5.241	-1.967	-5.409	-8.300	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	90.455
2-02-19	0	41.967	-1.617	-0.851	8.837	-0.370	3.000	3.000	1.000	0.000	0.000	0.000	0.022	278	-28.00	-44.284	-9.207	-1.103	-7.799	-7.347	-7.517	-7.799	-1.700	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	90.283
3-03-18	0	98.537	-0.290	-0.556	0.052	-2.957	4.000	4.000	1.000	0.000	0.000	0.000	0.150	424	-76.00	-4.945	-19.999	-8.197	-3.713	-5.635	-5.426	-3.713	-2.005	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	90.093
6-06-16	0	44.352	-2.158	-1.606	0.669	-3.457	10.000	10.000	1.000	0.000	-38.80	-38.491	-1.679	903	-118.00	-27.316	-9.530	-7.880	-9.138	-4.472	-3.060	-9.138	-9.420	0.00	5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	89.936
3-03-03	0	75.216	-0.049	-5.200	7.862	-8.914	4.000	4.000	1.000	0.000	0.000	0.000	-0.165	325	-96.00	-0.908	-16.661	-7.987	-8.854	-7.369	-7.177	-8.854	-3.455	0.00	2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	88.811
2-02-19	0	68.836	-6.902	-2.857	1.000	-4.567	2.000	3.000	1.000	0.000	0.000	0.000	-9.883	302	-40.00	-41.548	-8.896	-7.303	-30.25	-31.80	-8.608	-30.25	-11.205	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	87.829
2-02-19	0	62.153	-5.883	-8.004	4.000	-3.067	2.000	4.000	1.000	0.000	0.000	0.000	0.899	284	-14.00	-44.084	-64.758	-1.636	-4.166	-9.138	-7.408	-4.166	-8.381	0.00	1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	86.724

11. Tail error diagnostics (top 10% abs-error) (2/2)

Date	ted_diff	assey	efvol	volatility	volatility	gainyesgap	gap_pa	lamendcomp	comp_pa	Tfcost	Stency	row_indy	tred_pred	pred_epred	pred_hpred	pred_gpred	pred_xhistp	pred_qbtreft	pred_rawd	raw_cabad	sqrd_fraprdsi	proces	abs_error
-06-27	028.088	-4.515	13.122	29.974	42.180	10.000	1.000	0.000	1.000	18.69	-17.21	-0.354	928	-96.00	-8.188	11.40	-38.67	12.83	15.33	10.48	12.83	86.130	35.765
-06-16	017.19	-28.50	-42.00	34.46	34.372	10.000	0.000	1.000	0.000	15.40	15.84	-0.336	899	-126.06	28.47	-52.17	-21.51	-55.16	-57.05	-54.92	-55.16	47.955	34.940
-01-24	060.172	10.461	5.429	45.004	45.000	1.000	0.000	0.000	1.000	0.000	0.000	0.826	176	112.00	16.425	7.725	35.071	8.062	1.209	9.526	8.062	22.440	34.929
-02-18	075.033	4.465	-5.750	38.782	23.795	3.000	3.000	0.000	1.000	0.000	0.000	-1.549	258	82.000	20.03	-26.05	-3.592	32.86	38.06	39.32	32.86	36.590	34.737
-06-12	00.215	-3.715	13.893	33.757	49.379	11.000	9.000	0.000	1.000	27.94	28.200	-0.310	861	-84.00	6.562	-4.444	0.639	-8.389	-8.672	-5.841	-8.389	1.536	34.507
-04-16	00.711	8.681	4.667	25.832	0.486	6.000	6.000	1.000	0.000	11.73	-11.66	-1.529	549	-40.00	39.253	1.366	1.794	4.536	0.992	2.169	4.536	3.236	33.649
-03-18	060.698	2.425	8.306	13.952	5.826	3.000	4.000	1.000	0.000	0.000	0.000	-1.559	428	90.000	-1.724	6.478	3.284	-1.645	-2.969	-3.253	-1.645	5.468	33.241
-05-30	005.84	8.703	6.800	36.047	23.357	9.000	9.000	1.000	0.000	-3.733	-3.577	-0.087	817	92.000	1.665	7.411	0.186	7.515	6.971	9.674	7.515	4.450	32.915
-04-14	02.881	-8.391	11.332	6.692	4.187	6.000	6.000	0.000	1.000	8.267	8.165	0.678	531	44.000	17.60	-23.09	-37.02	-48.80	-39.54	-40.00	-48.80	-15.200	32.489
-03-18	093.865	5.417	-0.036	46.208	0.148	6.000	5.000	1.000	0.000	17.26	-17.22	0.122	405	84.000	5.616	-2.884	28.21	-2.646	0.167	-4.809	-2.646	3.868	32.358
-05-30	089.724	5.18	19.895	4.522	6.609	1.000	0.000	0.000	1.000	-3.582	-3.886	0.377	829	-114.06	32.07	-31.15	-43.32	-49.48	-52.03	-40.64	-49.48	35.660	32.008
-03-04	083.162	2.738	1.378	26.949	5.477	3.000	4.000	1.000	0.000	0.000	0.000	-1.635	382	70.000	-7.414	3.189	20.57	-19.63	-18.17	-18.92	-19.63	-9.645	31.780
-02-05	083.68	6.332	5.500	72.004	5.000	2.000	1.000	1.000	0.000	0.000	0.000	0.569	249	44.000	74.455	95.117	6.227	4.877	8.348	1.660	4.877	5.400	31.611
-04-30	069.000	5.75	6.002	5.158	4.677	8.000	8.000	1.000	0.000	23.25	-22.79	-0.643	629	62.000	13.044	1.53	-6.075	0.50	-8.48	-4.66	0.50	-6.990	31.418
-02-19	03.837	-0.798	-8.000	0.623	9.708	3.000	4.000	1.000	0.000	0.000	0.000	0.717	267	-112.00	0.202	6.601	-5.70	9.33	-1.36	-6.61	9.33	-7.200	31.088
-04-16	009.84	1.972	2.923	0.509	1.519	7.000	5.000	1.000	0.000	5.886	5.665	0.466	574	-22.000	0.907	7.326	1.485	3.29	2.326	1.078	3.294	7.320	30.867
-04-02	038.162	0.992	0.156	3.276	2.566	5.000	5.000	1.000	0.000	0.000	0.000	1.421	520	-38.000	2.943	9.933	0.169	4.28	1.27	8.257	4.28	6.540	30.701
-04-02	062.369	4.158	5.252	6.683	4.573	5.000	7.000	1.000	0.000	-0.743	1.031	-0.032	513	-62.000	1.996	1.489	8.958	8.814	0.857	3.625	8.814	2.110	30.674

11. Residual diagnostics plots



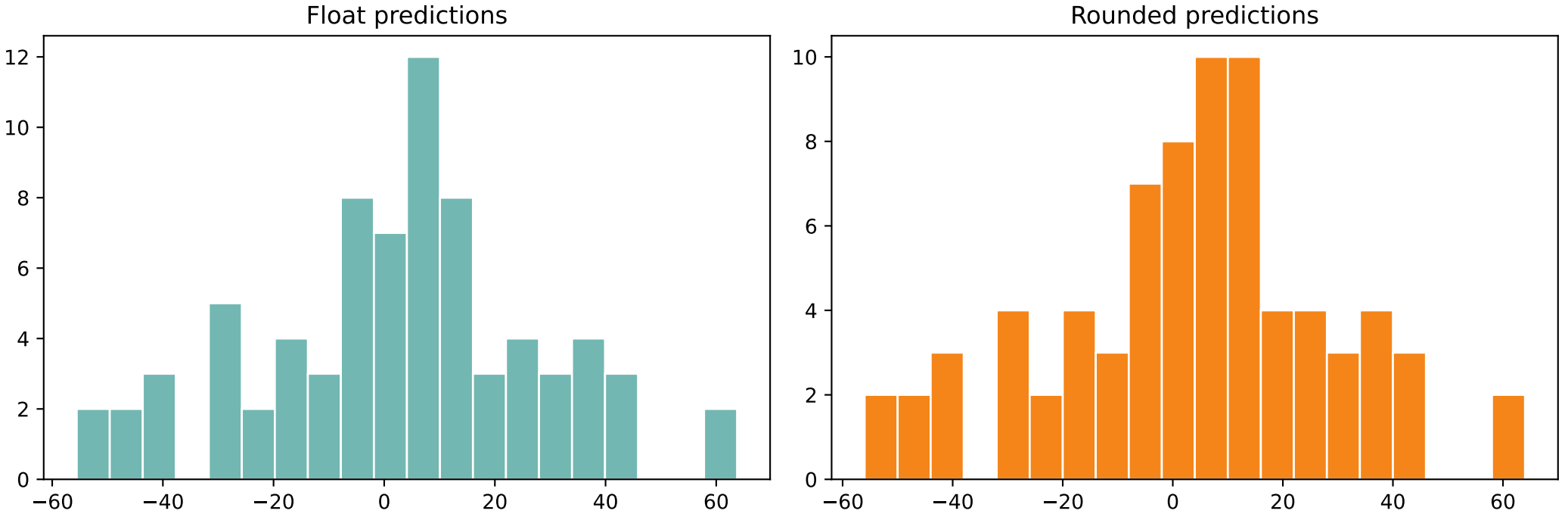
12-14. Final Decisions / Limitations / Next Steps

```
12. Final derby prediction distribution
- count=75, mean=2.733, std=25.265
- min=-55.569, max=63.771
- quantiles 1/5/25/50/75/95/99 = -55.376, -41.812, -10.708, 5.240, 16.474, 41.473, 60.080

13. Final decisions and rationale
- Final core candidate: elo_dynamic_k_teamha_decay|signal_core|hl=none|ridge=4|hgb=0
- Final postprocess candidate: module=expand_affine_global|stack=pool|cal=none|scale=none|q50=0.00|win=0.000|eta=0.55
- Recency profile=signal_core | half_life=None
- Shift mitigation applied=True
- Dispersion guard applied=False

14. Limitations + next steps
- Derby labels hidden: final selection relies on nested train-only proxies.
- No injuries/lineups/travel features available.
- Future work: monotone piecewise regime calibration and richer Bayesian uncertainty.
```

12. Final derby prediction distribution



Appendix A. First 10 rows of predictions.csv (1/1)

GameID	Date	Team1_Conf	Team1_ID	Team1	Team2_Conf	Team2_ID	Team2	Team1_WinMargin
1941	2025-07-04	Purple	68	Idaho	Yellow	119	Houston	-1
1942	2025-07-04	White	105	Helena	Yellow	120	Iowa	23
1943	2025-07-04	Red	82	Jackson	Purple	67	Concord	0
1944	2025-07-04	White	108	Montpelier	Yellow	129	West Virginia	4
1945	2025-07-04	Yellow	121	Lansing	White	113	South Dakota	-16
1946	2025-07-04	Crimson	21	Lincoln	Yellow	118	Florida	17
1947	2025-07-04	Yellow	117	Columbia	Crimson	23	Nashville	1
1948	2025-07-04	Orange	63	New Jersey	Yellow	126	Pierre	27
1949	2025-07-04	Purple	77	Wilmington	Orange	58	Honolulu	5
1950	2025-07-04	Crimson	22	Michigan	Green	49	Natchez	15

Appendix B. Top 20 rankings by Rank (1/1)

TeamID	Team	Rank
36	Massachusetts	1
94	Kentucky	2
90	Big Sur	3
37	Minnesota	4
45	Boise City	5
153	Tokyo	6
40	Pennsylvania	7
2	Baton Rouge	8
26	Phoenix	9
92	Illinois	10
88	Port St. Lucie	11
137	Milan	12
46	Boston	13
54	Boise	14
19	Georgia	15
47	Chicago	16
13	Texas	17
149	Seoul	18
131	Belgrade	19
116	California	20

Appendix C. Selected hyperparameters (1/1)

name	value
elo_variant	elo_dynamic_k_teamha_decay
feature_profile	signal_core
half_life_days	none
ridge_alpha	4.0
histgb_idx	0
histgb_params	{'learning_rate': 0.04, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10, 'l2_regularization': 0.3, 'ma
postprocess	module=expand_affine_global|stack=pool|cal=none|scale=none|q50=0.00|win=0.000|eta=0.55
postprocess_module	expand_affine_global
postprocess_eta	0.55
winsor_bounds	(-116.61, 113.305)
dispersion_guard_factor	1.0
seed	23
OMP_NUM_THREADS	1
MKL_NUM_THREADS	1

Appendix D. Final derby stage outputs (head) (1/1)

Date	elo_diff	passes	of_def	volatility	volatility	relative	saves	payer	payer	contender	contender	contender	consistency	consistency	row_index	pred_prob	pred_emp	pred_hum	pred_hist	pred_gr	pred_qr	pred_hist	pred_qr	pred_qr	stage8_s	stage_sa	stage_sa	stage_ca	stage_ca	stage_ca	stage_ca	stage_fir	spread_q
-07-04 0	15.692	-0.364	-3.797	35.596	29.274	11.000	12.000	0.000	1.000	11.839	11.230	-0.229	0	5.598	-3.070	40.59	-2.820	-1.618	-6.064	-2.820	3.244	-22.39	22.243	-1.145	-1.145	-1.145	-1.145	-1.337	44.637
-07-04 0	-0.877	14.106	12.458	39.164	40.902	10.000	12.000	0.000	1.000	-20.806	-20.935	0.499	1	27.310	23.342	40.59	10.484	9.230	1.894	10.484	8.189	-0.915	35.743	23.109	23.109	23.109	23.109	22.917	36.658
-07-04 0	31.326	-3.146	-7.864	59.498	27.316	11.000	13.000	0.000	1.000	9.849	10.542	-0.544	2	15.094	-5.894	40.59	13.289	9.165	21.596	13.289	-1.590	21.172	8.581	0.660	0.660	0.660	0.660	0.468	49.754
-07-04 0	13.783	2.010	-4.412	44.919	28.756	10.000	11.000	0.000	1.000	21.600	20.617	0.109	3	11.874	1.137	40.59	9.050	6.051	4.468	9.050	2.418	-26.09	28.301	3.939	3.939	3.939	3.939	3.747	54.395
-07-04 0	26.765	10.87	-16.75	38.940	36.984	12.000	10.000	0.000	1.000	-5.000	-4.714	-0.546	4	-4.545	18.811	40.59	-14.30	12.746	24.834	-14.30	18.023	-41.76	9.128	-15.66	-15.66	-15.66	-15.66	-15.856	60.896
-07-04 0	10.851	8.446	1.765	30.409	33.738	11.000	11.000	0.000	1.000	-21.52	-21.281	0.054	5	27.399	14.231	40.59	16.427	17.994	15.475	16.427	19.739	-1.111	40.331	17.459	17.459	17.459	17.459	17.267	41.442
-07-04 0	8.147	1.343	2.118	40.055	39.435	11.000	11.000	0.000	1.000	-2.073	-2.498	-0.082	6	2.759	0.591	40.59	6.084	7.109	9.128	6.084	3.678	-21.05	17.045	1.583	1.583	1.583	1.583	1.391	38.098
-07-04 0	-5.594	14.158	8.985	35.920	32.824	10.000	11.000	0.000	1.000	7.418	7.952	0.381	7	37.540	23.817	40.59	17.779	20.737	20.366	17.779	14.972	3.062	38.345	26.872	26.872	26.872	26.872	26.679	35.282
-07-04 0	51.907	0.807	-0.588	41.956	41.415	11.000	11.000	0.000	1.000	-17.09	-17.161	-0.023	8	15.198	2.882	40.59	5.896	-0.068	-7.331	5.896	-0.533	-19.91	19.155	5.344	5.344	5.344	5.344	5.152	39.065
-07-04 0	19.655	13.143	21.640	28.163	37.988	10.000	11.000	0.000	1.000	-12.92	-13.751	1.020	9	-0.332	20.788	40.59	12.516	8.844	6.395	12.516	8.179	-11.56	37.531	14.950	14.950	14.950	14.950	14.758	49.098
-07-04 0	16.139	6.094	1.544	41.287	34.893	10.000	11.000	0.000	1.000	-3.855	-3.763	0.003	10	15.825	8.337	40.59	12.178	8.646	9.066	12.178	12.752	-6.388	27.268	10.197	10.197	10.197	10.197	10.005	33.656
-07-04 0	11.881	-4.250	-8.824	32.105	38.441	11.000	11.000	0.000	1.000	-9.964	-9.633	-0.303	11	15.346	-8.158	40.59	-6.485	14.341	-4.522	-6.485	-2.650	-9.991	20.693	-2.345	-2.345	-2.345	-2.345	-2.537	30.684
-07-04 0	14.641	1.607	0.608	31.302	40.159	11.000	12.000	0.000	1.000	-0.827	-0.358	0.101	12	18.569	2.207	40.59	-11.67	13.426	-9.104	11.677	6.280	-8.493	22.017	5.432	5.432	5.432	5.432	5.240	30.510
-07-04 0	15.851	-2.432	-8.588	34.862	33.405	12.000	11.000	0.000	1.000	-12.904	-12.629	-0.039	13	-2.513	-4.629	40.59	-3.507	-2.649	-9.043	-3.507	-0.309	20.751	17.840	-4.333	-4.333	-4.333	-4.333	-4.525	38.592
-07-04 0	11.103	7.018	9.059	33.643	31.326	11.000	11.000	0.000	1.000	1.200	-0.057	0.401	14	-5.768	10.462	40.59	0.960	1.275	-1.836	0.960	-6.050	19.920	21.111	5.887	5.887	5.887	5.887	5.694	41.030
-07-04 0	12.821	0.733	-3.882	33.735	36.430	11.000	11.000	0.000	1.000	-19.606	-19.916	-0.155	15	-4.002	-0.706	40.59	8.815	-2.047	10.299	8.815	3.747	-15.91	34.321	-0.862	-0.862	-0.862	-0.862	-1.054	50.232
-07-04 0	11.062	5.344	-0.026	28.877	37.659	12.000	11.000	0.000	1.000	11.533	12.000	0.065	16	13.392	7.788	40.59	-2.346	-6.919	-2.754	-2.346	6.611	-16.07	20.411	8.513	8.513	8.513	8.513	8.321	36.486
-07-04 0	44.231	-0.458	-2.706	58.962	38.414	11.000	12.000	0.000	1.000	41.327	40.677	-0.193	17	-2.988	-4.917	40.59	-15.84	16.116	19.064	15.840	10.194	-41.11	5.352	-5.257	-5.257	-5.257	-5.257	-5.449	46.468
-07-04 0	15.534	6.648	-6.941	33.206	35.506	11.000	11.000	0.000	1.000	2.327	2.751	-0.150	18	35.956	11.289	40.59	26.177	22.783	27.130	26.177	21.126	2.281	35.886	18.105	18.105	18.105	18.105	17.912	33.605
-07-04 0	25.720	-3.761	-6.333	44.267	37.696	12.000	12.000	0.000	1.000	35.100	34.006	-0.273	19	-0.262	-9.450	40.59	-12.19	16.388	16.744	-12.19	12.029	-33.61	6.116	-7.641	-7.641	-7.641	-7.641	-7.834	39.733